

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 50

Annexure A

Scenario-Based

Code of Conduct:

I, D. Madhujithan ¹⁹²⁵²⁴²¹⁴ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Madhujithan

78/50

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

	Understanding of Scenario (20%)	Identification of Key issue (20%)	Application of Concepts (25%)	Decision making (20%)	Quality of Response (15%)
Q ₁	1.5	1.5	2	1	1
Q ₂	1	2	2	1	1
Q ₃	1.5	2	1.5	1.5	1.5
Q ₄	2	1	2.5	1.5	1.5
Q ₅	2	1.5	2	2	1

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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

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Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

I D.MADHUSUDHAN (192524214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

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RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1.5)	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification (1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

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4. Determine the remaining unused IP address range within the given network block.

Smart Campus Network — 192.168.100.0/24

The given network is 192.168.100.0/24, which contains 256 total IP addresses.

Subnet Allocation

Department	CIDR	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62

Department	CID R	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Remaining unused IP range:

The allocated addresses end at 192.168.100.223.

Therefore, the remaining unused range is:

192.168.100.224 – 192.168.100.255

This contains 32 IP addresses.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

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Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
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4. Identify the **unused IP address range** remaining within the allocated IP block.

Software Company Network — 10.10.0.0/16

The network 10.10.0.0/16 contains: $2^{16} = 65,536$ total IP addresses

Using the given subnet sizes sequentially:

Department	CIDR	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Unused IP address range:

The last allocated address is 10.10.1.223.

Therefore, the remaining unused range is:
10.10.1.224 – 10.10.255.255

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

Smart Hospital IPv6 Network

Given IPv6 address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed notation

IPv6 allows consecutive groups of zeros to be replaced by ::.

Therefore:

2001:db8:abcd::/64

2. Expanded hexadecimal representation

The full expanded form is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Network prefix

Since the prefix length is /64, the first 64 bits represent the network.

Therefore, the network prefix is:

2001:0db8:abcd:0000/64

or in compressed form:

2001:db8:abcd::/64

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

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192.168.3.0/24

A packet arrives at the router with the destination IP address:

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As the routing engineer, determine:

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Total host addresses
IPv6 has 128 bits.

With a /64 prefix:
Host bits = $128 - 64 = 64$ bits

Therefore:
 $2^{64} = 18,446,744,073,709,551,616$

5. Total possible host addresses

The subnet supports:
18,446,744,073,709,551,616 possible IPv6 addresses
Unlike IPv4, IPv6 does not normally subtract network and broadcast addresses, because IPv6 has no broadcast address.

4. ISP Routing Table

Routing table:

Routing Entry
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Destination IP:
192.168.2.45

1. Destination subnet
The destination IP 192.168.2.45 belongs to:
192.168.2.0/24
Because the /24 network covers:
192.168.2.0 – 192.168.2.255

2. Matching routing table entry
The matching entry is:
192.168.2.0/24

3. Next-hop network/interface
The router forwards the packet through the interface connected to the:
192.168.2.0/24 network

The exact next-hop IP/interface cannot be determined from the information given. It would depend on the router's actual interface configuration.

4. Default route

If no routing-table entry matches the destination, the router uses the default route:

0.0.0.0/0

The actual next-hop IP for this default route is not provided in the question.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
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Administration and Finance LANs

Given:

- Administration LAN → 192.168.10.0/26**
- Finance LAN → 192.168.10.64/26**

A /26 provides 64 total addresses and 62 usable host addresses.

Administration LAN

Network: 192.168.10.0/26

Item	Address
Network Address	192.168.10.0
First Valid Host	192.168.10.1

Item	Address
Last Valid Host	192.168.10.62
Broadcast Address	192.168.10.63
Usable Hosts	62
Default Gateway	192.168.10.1

Three suitable computer IP addresses

- Computer 1 → 192.168.10.2
- Computer 2 → 192.168.10.3
- Computer 3 → 192.168.10.4

All should use:

Default Gateway: 192.168.10.1

Finance LAN

Network: 192.168.10.64/26

Item	Address
Network Address	192.168.10.64
First Valid Host	192.168.10.65
Last Valid Host	192.168.10.126
Broadcast Address	192.168.10.127
Usable Hosts	62
Default Gateway	192.168.10.65

Three suitable computer IP addresses

- Computer 1 → 192.168.10.66
- Computer 2 → 192.168.10.67
- Computer 3 → 192.168.10.68

All should use:

Default Gateway: 192.168.10.65